
DATABASE MANAGEMENT SYSTEMS (IT 2040)

LECTURE 05 – DATABASE PROGRAMMING



LECTURE CONTENT

- Views
- SQL extensions (specifically T-SQL)
- Functions & Procedures
- Triggers

LEARNING OUTCOMES

- At the end of this lecture, students should be able to
 - Identify situations where views, functions, stored procedures and triggers are applicable.
 - Write syntactically correct sql statements to create functions, procedures and triggers on RDBMS to cater user requirements.

VIEWS

- A **view** is a virtual table which is derived from other tables which are called **base tables** or **defining tables**.
- A view does not necessarily exist in a physical form.
- In SQL, CREATE VIEW statement is used to define views.
- Syntax :

```
CREATE VIEW <view_name> AS  
SELECT <column_name(s)>  
FROM <table_name>  
WHERE <condition>
```

VIEWS (CONTD.)

- emp (eid, ename, age, salary) ,dept (did, dname, budget,mangerId), works (eid,did, pct_time)
- Create a view named dept_info that contains name of the department, budget and manager's name

```
CREATE VIEW dept_info(dname,budget,manager)
```

```
AS
```

```
SELECT d.dname, d.budget, e.ename FROM emp e, dept d
```

```
WHERE e.eid=d.managerId
```

EXERCISE I

- Create a view named emp_info which contains eid, name, salary and total percentage of time.

emp (eid, ename, age, salary)

dept (did, dname, budget, mangerId)

works (eid, did, pct_time)

QUERYING AND DELETING A VIEW

- Querying a view can be done similar to a table.
 - Ex : `SELECT * FROM dept_info`
 - Ex : `SELECT dname FROM dep_info WHERE budget > 500,000`
- Dropping a view
 - Ex : `DROP VIEW dept_info`

UPDATING VIEWS (CONTD.)

- Updating a view can be ambiguous...
 - Views containing aggregate functions are not updateable

- Ex :

```
UPDATE emp_info  
SET  tot_pct= 90  
WHERE eid = 1000
```

**This is not possible

UPDATING VIEWS (CONTD.)

- Views containing a join can be ambiguous

A	B
b	1

B	C
1	d
1	e

A	B	C
b	1	d
b	1	e

UPDATE VI

SET A = a

WHERE C = e

- Thus, in many DBMSs, views are updateable only if they are defined on a single base table.

WHY VIEWS ?

■ Advantages

- **Security** : Each user can be given permission to access the database only through a small set of views that contain the specific data the user is authorized to see, thus restricting the user's access to stored data
- **Query Simplicity** : A view can draw data from several different tables and present it as a single table, turning multi-table queries into single-table queries against the view.

■ Disadvantages

- **Performance** : Views create the appearance of a table, but the DBMS must still translate queries against the view into queries against the underlying source tables. If the view is defined by a complex, multi-table query then simple queries on the views may take considerable time
- **Update restrictions**

PROGRAMMING IN T-SQL

- Similar to a programming language, certain extensions have been made in SQL to program simple server-side logic.
- Some of the statements include:
 - Variables
 - Selection conditions
 - IF (...)... ELSE ...
 - Looping
 - WHILE (...)

T-SQL: VARIABLES

- A Transact-SQL local variable is an object that can hold a single data value of a specific type.
- Variables in scripts are typically used:
 - As a counter either to count the number of times a loop is performed or to control how many times the loop is performed
 - To hold a data value to be tested by a control-of-flow statement
 - To save a data value to be returned by a stored procedure return code.

T-SQL: VARIABLES (CONTD.)

- The DECLARE statement initializes a Transact-SQL variable.
 - Syntax: DECLARE @<variable name> <data type>
 - Ex: DECLARE @DName VARCHAR(20)
 - The created variable will be holding a null value
- To assign a value to a variable, use the SET statement.
 - Syntax : SET @<variable name> =<value>
 - SET @DName = 'SESD'

T-SQL: VARIABLES (CONTD.)

- The declared variables could be used in scripts
- Ex :
 - ```
SELECT budget
FROM Dept
WHERE dname = @DName
```
  - ```
DECLARE @empld INT
SELECT @empld = MAX(eid)
FROM emp
```

T-SQL: IF STATEMENT

- Imposes conditions on the execution of a Transact-SQL statement.

- Ex:

```
IF (SELECT count(eid) FROM emp) > 1000
```

```
BEGIN
```

```
    PRINT 'Inside the IF statement'
```

```
    PRINT 'There are lesser than 1000 employees '
```

```
END
```

```
ELSE
```

```
    PRINT 'There are more than 1000 employees ! '
```

T-SQL: WHILE STATEMENT

- Sets a condition for the repeated execution of an SQL statement or statement block.
- The statements are executed repeatedly as long as the specified condition is true.
- The execution of statements in the WHILE loop can be controlled from inside the loop with the BREAK and CONTINUE keywords.

T-SQL: WHILE STATEMENT (CONTD.)

- BREAK

- Causes an exit from the innermost WHILE loop. Any statements appearing after the END keyword, marking the end of the loop, are executed.

- CONTINUE

- Causes the WHILE loop to restart, ignoring any statements after the CONTINUE keyword.

T-SQL: WHILE STATEMENT (CONTD.)

- Ex :

```
WHILE @count <= 100
```

```
BEGIN
```

```
    INSERT INTO Employees VALUES (@count, CONCAT('Employee', @count))
```

```
    SET @count = @count + 1
```

```
END
```

STORED FUNCTIONS/PROCEDURES

- Business logic is maintained in database tier for data intensive operations
 - E.g. Calculating all interest earned in bank accounts
- In SQL Server 2005, Stored Procedure/ Functions can be written in
 - T-SQL (we will study only this)
 - Any .NET Language

STORED FUNCTIONS/PROCEDURES (CONTD.)

- Syntax of a function

```
CREATE FUNCTION <function name> (parameters)  
RETURNS <return type>  
<function body>
```

- Parameter mode of parameters for functions is IN which parameters allow the calling code to pass values into the procedure

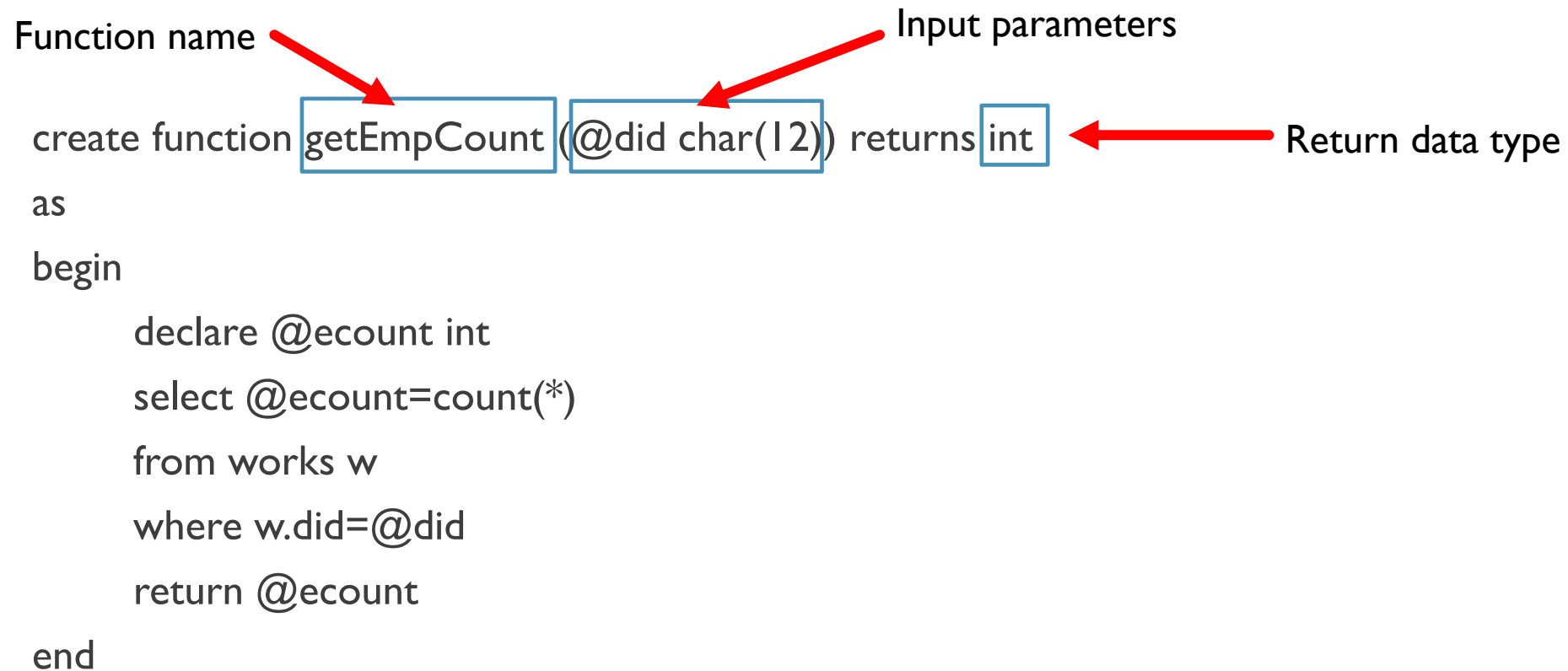
- Syntax of a procedure

```
CREATE PROCEDURE <procedure name> (parameters)  
<procedure body>
```

- Each parameter to a procedure should have a data type and a parameter mode (IN or OUT).
 - IN: This is the default mode. IN parameters allow the calling code to pass values into the procedure
 - OUT: OUT parameters allow the procedure to pass values back to the calling code

FUNCTIONS

- Ex : Create a function that returns the number of employees in a given department.



The diagram illustrates the SQL syntax for creating a function. It shows the following code snippet:

```
create function getEmpCount (@did char(12)) returns int
as
begin
    declare @ecount int
    select @ecount=count(*)
    from works w
    where w.did=@did
    return @ecount
end
```

Annotations with red arrows point to specific parts of the code:

- Function name**: Points to `getEmpCount`.
- Input parameters**: Points to `(@did char(12))`.
- Return data type**: Points to `returns int`.

FUNCTIONS (CONTD.)

- Calling the function created previously

```
declare @result int
```

```
exec @result=get_empCount 'Admin'
```

```
print @result
```

EXERCISE 2

- Create a function to return the total percentage of time a person works given the employee id.

STORED PROCEDURES

- Ex : Create a procedure to give a salary increment to all the employee by a given percentage from their existing salary

```
CREATE PROCEDURE increaseSalary (@pct float)
```

```
as
```

```
begin
```

```
    Update emp
```

```
    Set salary=salary+salary * (pct/100)
```

```
end;
```


STORED PROCEDURES (CONTD.)

- Calling the procedure

```
exec increaseSalary 10
```

STORED PROCEDURES (CONTD.)

- Ex 2: create a procedure that outputs statistics of salary (min, max) for a given department.

```
create procedure get_stats(@did varchar(12),@maxm real output,@minm real output)
as
begin
    select @maxm=max(e.salary),@minm=min(e.salary)    from dept d, works w, emp e
    where d.did=w.did and w.eid=e.eid and d.did=@did
end
```

STORED PROCEDURES (CONTD.)

- Calling the procedure

```
declare @max int,@min int
```

```
exec get_stats 'Admin', @max output,@min output
```

```
print @max
```

```
print @min
```

EXERCISE 3

- Create a procedure that outputs the name of the manager and his salary in a given department.

TRIGGERS

- Triggers are useful in enforcing business rules and data integrity.
- They are more powerful than general constraints.
- For example,
 - The employees salary is always less than his/her manager's salary

T-SQL:TRIGGERS

- A trigger is a special type of stored procedure that automatically takes effect when the data in a specified table is modified.
- A trigger is invoked in response to a
 - DDL statement (CREATE,ALTER etc.) or
 - DML statement (INSERT, UPDATE, or DELETE statement).‘

T-SQL: TRIGGER SYNTAX

- We will learn DML triggers...

- Syntax :

```
CREATE TRIGGER trigger_name
ON { table | view }
{
    { { FOR | AFTER | INSTEAD OF }
    { [ INSERT ] [ , ] [ UPDATE ] [ , ]
    [ DELETE ] }
    AS
        sql_statement [ ...n ]
    }
}
```

T-SQL: TRIGGER SYNTAX (CONTD.)

- FOR|AFTER

- AFTER specifies that the DML trigger is fired only when all operations specified in the triggering SQL statement have executed successfully.
- AFTER is the default when FOR is the only keyword specified.
- AFTER triggers cannot be defined on views.

- INSTEAD OF

- Specifies that the trigger is executed *instead of* the triggering SQL statement, thus overriding the actions of the triggering statements.
- Specifies At most, one INSTEAD OF trigger per INSERT, UPDATE, or DELETE statement can be defined on a table or view.

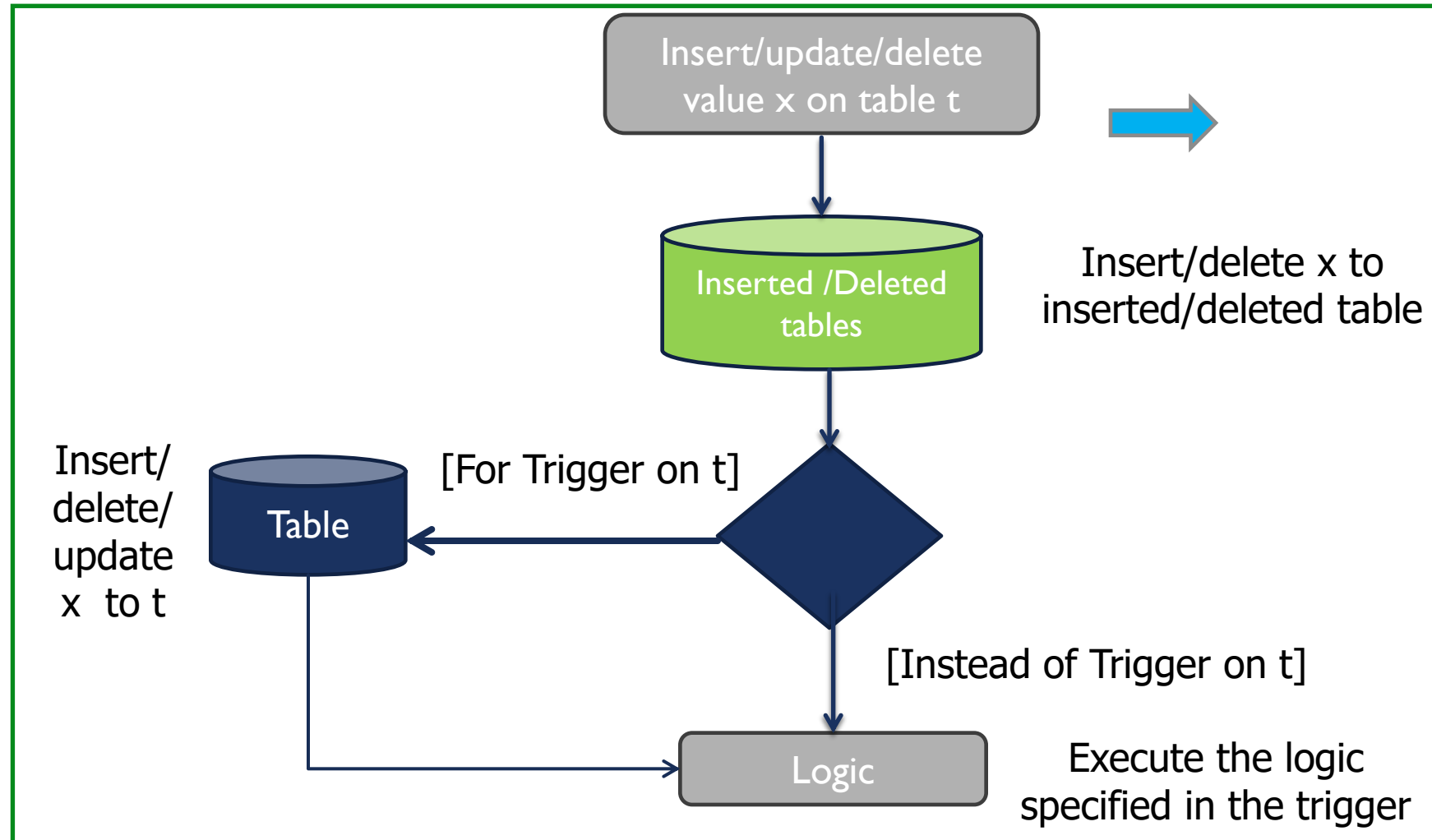
T-SQL: TRIGGER SYNTAX (CONTD.)

- [DELETE] [,] [INSERT] [,] [UPDATE]
 - Are keywords that specify which data modification statements, when attempted against this table or view, activate the trigger. At least one option must be specified.

TRIGGERS : INSERTED/DELETED TABLES

- When a trigger is executed SQL Server creates two virtual tables called INSERTED and DELETED.
- The **deleted** table stores copies of the affected rows during DELETE and UPDATE statements
- The **inserted** table stores copies of the affected rows during INSERT and UPDATE statements
 - For example when inserting a record to a table, SQL Server creates a virtual table call INSERTED and loads data into the inserted table then executes the trigger statements and writes the related data pages.
- The format of the inserted and deleted tables is the same as the format of the table on which the trigger is defined
- Each column in the inserted and deleted tables maps directly to a column in the base table

TRIGGERS : HOW DO THEY WORK?



T-SQL: TRIGGERS (CONTD.)

- Example 1:
 - Consider tables below
 - Account (accountNo, custId, branch, balance)
 - AccountAudit (accountNo, balance, date)
 - Create a trigger to track all inserts/updates done to the balance field of an Account table at a bank in the AccountAudit table

T-SQL: TRIGGERS (CONTD.)

Create trigger account_audit_trigg

On Account

For Insert, update

As

Begin

Declare @ano int

Declare @balance float

Select @ano=accountNo,@balance=balance from inserted

Insert into accountAudit(@ano,@balance,getdate())

end

T-SQL: TRIGGERS (CONTD.)

- Example 2:
 - Consider following tables :
 - Emp(eid ,ename, age, salary)
 - Works (eid, did, pct-time)
 - Dept(did, budget, managerid)
 - Create a trigger to ensure that an employee doesn't work in more than 2 departments

EXERCISE 4

- Consider the following table
 - Transaction(tid, accountNo, type, amount, date)
 - Type may contain 'credit' or 'debit'
- Assuming that the bank's maximum withdrawal limit per day is 40000, write a trigger to ensure that no customer withdraws more than the given limit.

EXERCISE 5

- Consider the tables given below
 - Employee(nic, name, salary, dno)
 - Dept (dno, dname, mgrNic)
- Create a trigger to ensure that no employee has a salary greater than his/her manager.

SUMMARY

- Views
- Transaction Basics
- T-SQL extensions
- Stored Procedures
- Triggers